

Aaptha Boggaram

Charlotte, NC | aaptha.boggaram@gmail.com | (303) 210-2242 | linkedin.com/in/aaptha-boggaram

EDUCATION

Master of Science in Computer Science , University of Colorado Boulder	May 2025
Bachelor of Technology in Computer Science & Engineering , PES University	May 2023

WORK EXPERIENCE

University of Colorado Boulder <i>Lead Teaching Assistant Boulder, CO</i>	Aug 2024 - May 2025
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- Conducted weekly coding labs on R programming concepts and data analysis for 50+ students and developed automated grading scripts in Python with test cases to streamline assignment evaluation.

The Kroger Co. <i>Software Engineering Intern Cincinnati, OH</i>	May 2024 - Aug 2024
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- Developed software independently to automate the creation of product variant groups, leveraging Large Language Models (LLMs) and Generative AI (GenAI) for efficient product information categorization, making it accessible to over **11 million Kroger users**.
- Designed and implemented major database changes for an upcoming release and worked closely with 84.51 to integrate the changes into a workflow to automate manual processes carried out through the merchant portal on Kroger's web application.
- Made improvements on backend features using Golang, including optimizing REST APIs for efficient data communication.
- Worked in an agile team, contributing to UI enhancements for Kroger's web application using React with Typescript.

Continental AG <i>Software Intern Bengaluru, India</i>	Jan 2023 - Jul 2023
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- Engaged in applied research as a core developer and implemented novel algorithms for road type classification and curve estimation for Continental's Advanced Radar Sensors, outperforming existing methods by 18% and 25%, respectively.
- Synthesized structured datasets from raw radar signals, generating 5+ high-fidelity datasets for downstream modeling and evaluation.
- Authored and submitted a peer-reviewed conference paper (IEEE CONECCT 2024) detailing methodology and results as the primary contributor.

LiVNSense Technologies <i>Summer Intern Remote</i>	Jun 2021 - Sept 2021
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- Developed and deployed a real-time computer vision framework for pothole detection using Python and C++ on an Arduino Nano BLE Sense, achieving a 94.5% reduction in model size via quantization while preserving high detection accuracy.

SKILLS

Languages: Python, C++, Java, C, SQL, Go (Golang), Typescript, JavaScript, R, PostgreSQL, Bash, HTML, CSS
Tools: Git, Docker, Kubernetes, AWS, Microsoft Azure, Gradle, Figma, Jira, Confluence, CI/CD, CMake, Spark, Hadoop, Spring
Frameworks & Libraries: React, TensorFlow, PyTorch, Flask, Scikit-Learn, Keras, Kafka, Pandas, NumPy, OpenCV, ROS2

PROJECTS

DotDNN: A Block-Grained Scaling Framework for Identifying the Best DNNs for Edge Vision
Developed a dynamic, block-grained scaling system enabling adaptive DNN selection and deployment on edge devices while maintaining optimal accuracy-latency trade-offs under fluctuating compute and memory constraints.

Offline Reinforcement Learning for Cardiac Pacemakers
Project done in collaboration with CU Anschutz. Developed a state-of-the-art offline reinforcement learning model, utilizing historical patient data from Medtronic pacemakers to optimize pacemaker parameters for improved cardiac rhythm pacing.

Cruella: Autonomous Vehicle Navigation System
Developed an autonomous vehicle controller using ROS2, implementing LIDAR-based SLAM for visual navigation, reverse parking, obstacle avoidance, and stop sign detection, recognized as one of the fastest teams to navigate a closed indoor course in under 2 minutes.

RealSign: Real-time bidirectional Indian Sign Language Translator
Led a team of 4 to develop an app for real-time bidirectional Indian Sign Language translation, training a Siamese Neural Network on a self-collated dataset of 34,000+ images, now deployed as one of the largest open-source ISL datasets.